Diagnostic Potential of EEG Theta Power and Interhemispheric Correlation of Temporal Lobe Activities in Alzheimer's Disease through Random Forest Analysis

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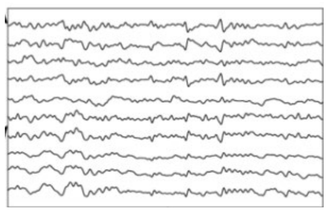
Introduction

Given the prevalence of Alzheimer's Disease (AD) among the elderly and the absence of a cure[1,2], early detection emerges as a critical focus area while electroencephalography (EEG) offers a promising diagnostic tool[3]. To date, several studies using EEG datasets have revealed high accuracy yet not exploring which features play a crucial role in distinguishing AD from healthy controls (HC) due to the black box nature of machine learning models[4,5,6,7]. This study explores the diagnostic capabilities of EEG on distinguishing AD from HC through random forest classification (RFC) and inspection of distribution of EEG features.

Methods

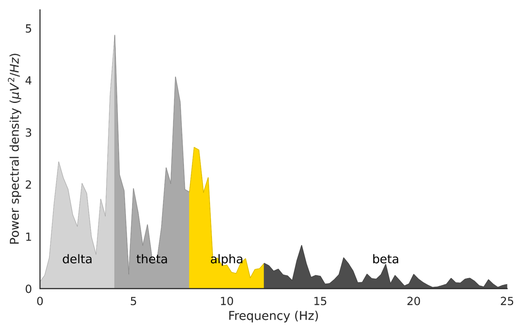
▪ Preprocessing

Resting state EEG



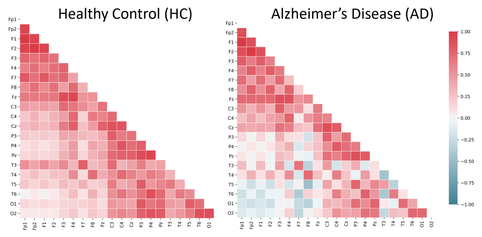
Duration : **8 sec**
Sampling frequency : **128 Hz**
21 Scalp Electrode
based on 10-20 System
Band range : **0.5Hz - 30Hz**
Movement artifacts removed
Dataset available through
https://osf.io/2v5md/?view_only=
(Vicchietti et al., 2023)[10]

Band Power (Welch, 1967)



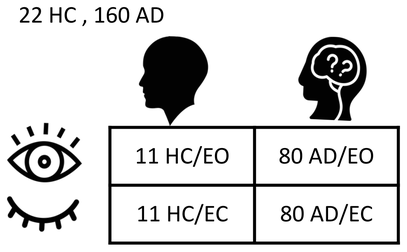
Cross Correlation

Pearson's correlation (Benesty et al., 2009)



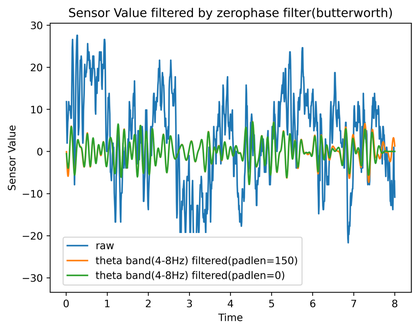
▪ Participants

4 Groups
:2 (**Health Condition**: AD/HC)* 2 (**Eye Open**: EO/EC)
22 HC , 160 AD



11 HC/EO	80 AD/EO
11 HC/EC	80 AD/EC

Theta band filtered Cross Correlation



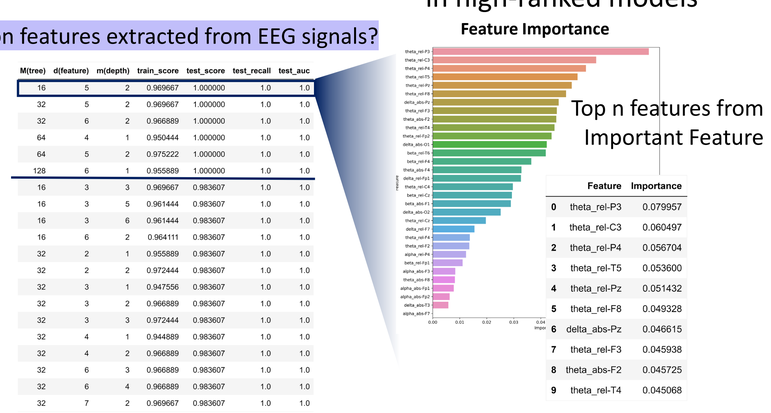
▪ Random Forest Classifier

1. **Fit** a Random Forest Classifier
2. **Sort** models by test score, recall, AUC
3. Manual **Selection** of **Important Features** in high-ranked models
4. **Visual Inspection** of Important Features Across models

Main Question :
Can **RFC** discriminate **AD and HC** based on features extracted from EEG signals?
Which features play a significant role?

Hyperparameters	Values
n_estimators	[2,4,8,16,32, 64,128,256]
bootstrap	True
class_weight	True
criterion	Gini
max_depth	8
max_features	8
max_leaf_nodes	8
max_samples	None

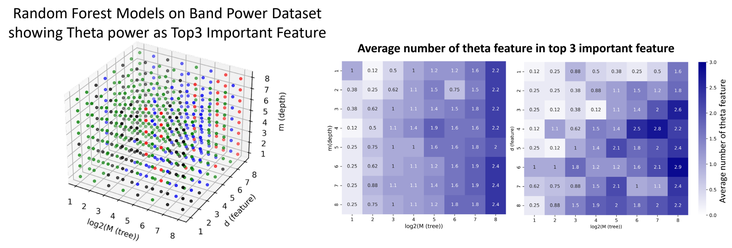
- 5-fold cross validation with n_repeats = 3
- Feature scaled by Min-max normalization
- Feature Importance analysis by mean decrease in impurity



M(tree)	d(feature)	m(depth)	train_score	test_score	test_recall	test_auc
16	5	2	0.969667	1.000000	1.0	1.0
32	5	2	0.969667	1.000000	1.0	1.0
32	6	2	0.966889	1.000000	1.0	1.0
64	4	1	0.950444	1.000000	1.0	1.0
64	5	2	0.975222	1.000000	1.0	1.0
128	6	1	0.955889	1.000000	1.0	1.0
16	3	3	0.969667	0.983607	1.0	1.0
16	3	5	0.961444	0.983607	1.0	1.0
16	3	6	0.961444	0.983607	1.0	1.0
16	6	2	0.964111	0.983607	1.0	1.0
32	2	1	0.955889	0.983607	1.0	1.0
32	2	2	0.972444	0.983607	1.0	1.0
32	3	1	0.947556	0.983607	1.0	1.0
32	3	2	0.966889	0.983607	1.0	1.0
32	3	3	0.972444	0.983607	1.0	1.0
32	4	1	0.944889	0.983607	1.0	1.0
32	4	2	0.966889	0.983607	1.0	1.0
32	6	3	0.966889	0.983607	1.0	1.0
32	6	4	0.966889	0.983607	1.0	1.0
32	7	2	0.969667	0.983607	1.0	1.0

	Feature	Importance
0	theta_rel-P3	0.079957
1	theta_rel-C3	0.060497
2	theta_rel-P4	0.056704
3	theta_rel-T5	0.053600
4	theta_rel-Pz	0.051432
5	theta_rel-F8	0.049328
6	delta_abs-Pz	0.046615
7	theta_rel-F3	0.045938
8	theta_abs-F2	0.045725
9	theta_rel-T4	0.045068

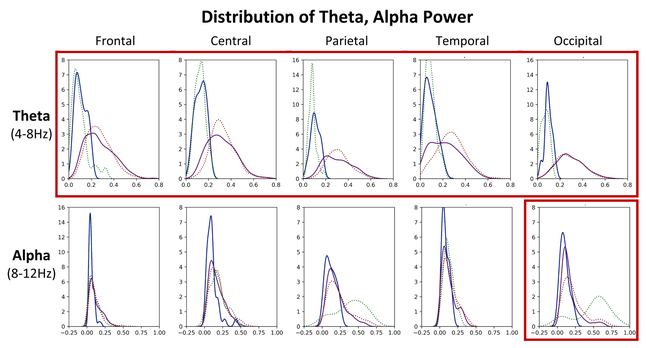
Random Forest Models on Band Power Dataset showing Theta power as Top3 Important Feature



Results

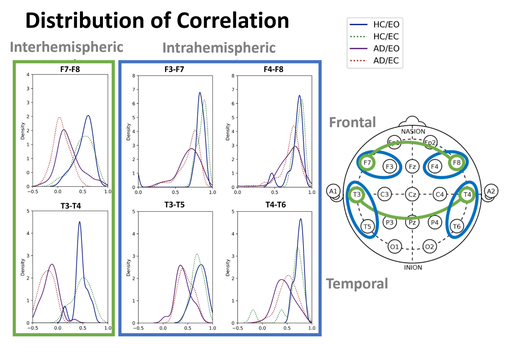
▪ Distribution plot

Band Power



Higher **Theta** power in all regions of AD patients (vs.HC)
Higher **Alpha** power in **Occipital** region of HC/EC group

Cross Correlation



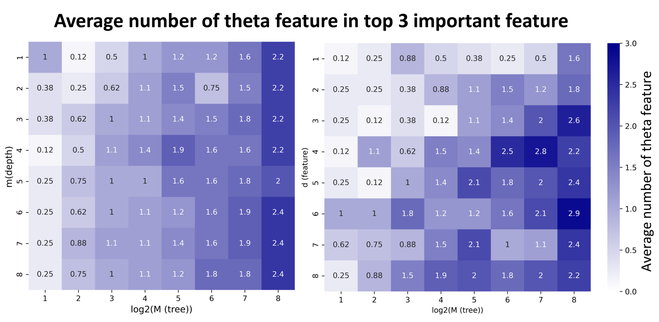
Distinguishable pattern in **Interhemispheric** channel pair of the **Frontal & Temporal** lobe

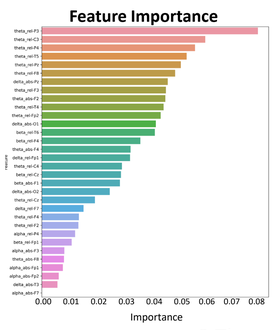
▪ Random Forest Classifier

Band Power

Theta feature

M(tree)	d(feature)	m(depth)	train score	test score	test recall	test AUC
16	5	2	0.97	1.0	1.0	1.0





	Feature	Importance
0	theta_rel-P3	0.079957
1	theta_rel-C3	0.060497
2	theta_rel-P4	0.056704
3	theta_rel-T5	0.053600
4	theta_rel-Pz	0.051432
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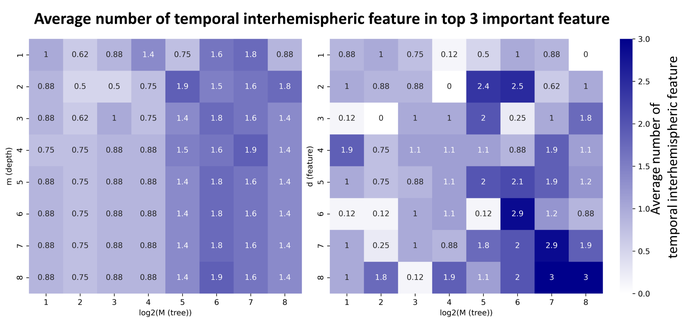
Theta feature was important across RFC models

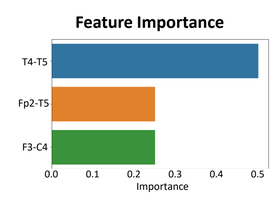
8 Theta feature
Included in top10 important features

Cross Correlation

Temporal Interhemispheric feature

M(tree)	d(feature)	m(depth)	train score	test score	test recall	test AUC
4	6	1	0.97	1.0	1.0	1.0





	Feature	Importance
0	T4-T5	0.50
1	F3-C4	0.25
2	Fp2-T5	0.25

Temporal interhemispheric feature was important across RFC models

2 Temporal feature
3 Interhemispheric feature
Included in top 3 important features

Theta band filtered Cross Correlation

Theta feature + **Temporal** feature

51 Models achieved perfect score
With only **6** features

Conclusion

- The highest ranked models in each dataset achieved 100% test accuracy, recall and AUC, though this extraordinarily high accuracy warrants caution due to the small dataset size with high data imbalance between groups, HC and AD.
- Important features in RFC models were in align with the distribution of the features within the dataset and findings in previous studies which denoted theta band being affected in the early stage of AD[8] and temporal lobes as parts of the brain with beta-amyloid deposition[9].
- This methodology demonstrates the efficacy of EEG-based metrics and machine learning in enhancing our understanding of EEG characteristic in AD patients and its diagnostic potential.

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Reference

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